

PENETRATION TESTING REPORT

FOOTPRINTING AND NETWORK DISCOVERY

NETWORKWALKS | CYBERSECURITY PROGRAMME | WEEK 2

Report information	Details
Company	Networkwalks
Report type	Educational penetration-testing and reconnaissance report
Modules completed	W2-PM1: Footprinting with multiple Kali tools; W2-PM5: Network scanning with Zenmap
External target	networkwalks.com
Internal target	Authorised local LAN owned or controlled by the student
Assessment focus	Public information discovery, web fingerprinting, DNS analysis, HTTP inspection, WAF identification, and local host discovery
Assessment date	19 August 2026
Submitted by	Cybersecurity Intern, Saad Kashif

Scope statement

The activities documented here were limited to the assigned educational targets and the author's own local network. No exploitation, credential attacks, persistence, or denial-of-service activity was performed.

CONFIDENTIALITY NOTE

The local-network addresses and hardware identifiers in the Zenmap evidence have been redacted in this report copy for privacy. The original screenshots should be retained separately for private academic submission.

Executive Summary

This report records two controlled cybersecurity exercises completed during Week 2 of the programme. The first exercise examined the public footprint of networkwalks.com through six reconnaissance tools in Kali Linux. The second exercise used Zenmap on a local Windows environment to identify live hosts within an authorised private subnet and produce a network topology view.

The footprinting activity established a clear picture of the target's registration, hosting, web technology, DNS, HTTP, and protective-control characteristics. The evidence identified GoDaddy as the registrar, HostGator name servers, an Apache and WordPress-based web presence, a public web address, a ModSecurity Web Application Firewall, and DNS records associated with hosting and mail services.

The Zenmap exercise identified four live hosts on the scanned /24 subnet. The submitted screenshots show a successful ping scan and a topology view with the local devices represented graphically. The local IP addresses and MAC addresses are intentionally not reproduced in this public-facing report version.

Overall assessment

The exercises did not confirm a compromise or a vulnerability. They demonstrated how much infrastructure context can be collected before intrusive testing begins, and why organisations should reduce unnecessary technical disclosure and maintain an accurate inventory of their own networks.

Key Findings at a Glance

Area	Evidence-based observation	Assessment
Domain registration	GoDaddy registrar; registration created in 2019 and set to expire in 2027; privacy-protected registrant details.	Informational
Web technology	Apache, WordPress 7.0.4, WordPress Download Manager 3.3.58, Bootstrap 7.0.4, jQuery 3.7.1, and Google Tag Manager indicators.	Medium
DNS resolution	networkwalks.com resolved to 192.232.216.135 through Google Public DNS at 8.8.8.8.	Low
HTTP response	HTTP/2 200 response; Apache server banner; WordPress REST API references; secure and HttpOnly cookie observed.	Low
Web protection	wafw00f identified ModSecurity associated with SpiderLabs.	Informational
DNS footprint	HostGator name servers, mail routing, SPF/TXT records, cPanel autodiscover services, and BIND version indicators.	Medium
Local network discovery	Zenmap ping scan identified four live hosts on the authorised /24 subnet.	Medium

1. Authorisation, Ethics, and Limitations

All activity documented in this report was completed for an educational cybersecurity exercise. The public-domain work was performed against the assigned Networkwalks target, and the Zenmap work was performed against the author's own local network. The work was limited to reconnaissance and host discovery.

Reconnaissance results are observations rather than proof of weakness. A detected technology version may be inaccurate or may be protected by compensating controls. A live host is not necessarily vulnerable, and a visible DNS or HTTP record does not establish unauthorised access. Confirming a vulnerability would require a separate, explicitly authorised validation phase.

Handling requirement

Do not publish the original Zenmap screenshots with live IP addresses, MAC addresses, hostnames, or router details. Use the redacted report copy for public sharing and retain the original evidence for private submission.

2. Engagement Overview

Engagement element	Description
Objective	Document the public footprint of networkwalks.com and map live systems on an authorised local network.
Reconnaissance environment	Kali Linux terminal with WHOIS, WhatWeb, nslookup, curl, wafw00f, and DNSRecon.
Network discovery environment	Physical Windows PC running Zenmap, with a Ping Scan against the local /24 subnet.
Evidence collected	Command outputs, task screenshots, a Zenmap scan screenshot, and a Zenmap topology screenshot.
Excluded activity	Exploitation, password attacks, vulnerability validation, service abuse, and unauthorised scanning.

3. Methodology

The assessment followed a progressively richer information-gathering sequence. WHOIS established administrative context. WhatWeb and curl examined the public web stack. nslookup and DNSRecon mapped name resolution and published service records. wafw00f identified the apparent web application protection layer. Zenmap then moved from public-domain reconnaissance to controlled discovery of live hosts on the author's local network.

Assessment objective	Tool	Evidence collected
Administrative footprint	WHOIS	Registration, registrar, dates, status, and name servers.
Web fingerprinting	WhatWeb	Web server, CMS, plugins, libraries, redirects, and public metadata.
Address resolution	nslookup	Resolver used and the observed IPv4 address.
HTTP inspection	curl -I	Status, headers, cookie attributes, caching indicators, and linked endpoints.
Protection detection	wafw00f	WAF identification based on response signatures.
DNS enumeration	DNSRecon	SOA, NS, MX, TXT/SPF, SRV, and DNS software indicators.
Local discovery	Zenmap / Nmap	Live hosts, local addresses, MAC evidence, and topology view.

Evidence verification

All findings and numeric claims in this report were cross-checked against the supplied raw command outputs and screenshots before finalisation. Where the executed command differs from the normalised reporting form, both values are stated explicitly.

4. Footprinting and Reconnaissance Findings

The six footprinting tools produced complementary results. Taken together, they describe the target domain as a hosted WordPress site using Apache, with DNS and mail services associated with HostGator and cPanel infrastructure. The findings below are based on the saved command outputs supplied with the evidence package.

4.1 Domain Registration and Administrative Context

The WHOIS query identified GoDaddy.com, LLC as the registrar. The record shows a creation date of 6 November 2019 and a registry expiry date of 6 November 2027. Registrant details are privacy protected through Domains By Proxy, LLC. The domain uses HostGator name servers in the registry output and is marked with client-side transfer, update, renewal, and deletion restrictions.

```
whois networkwalks.com
```

WHOIS field	Observed result
Registrar	GoDaddy.com, LLC
Creation date	6 November 2019
Registry expiry	6 November 2027
Registrant visibility	Registration Private / Domains By Proxy, LLC
Name servers	NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM; additional registry entries also show NS29.DOMAINCONTROL.COM and NS30.DOMAINCONTROL.COM
DNSSEC	Unsigned
Domain status	Client delete, renew, transfer, and update prohibitions were observed

From a defensive perspective, the registration record shows that ownership details are not openly exposed, while the registrar, lifecycle dates, delegation, and administrative status remain visible. These are useful for asset management and should be monitored by the domain owner.

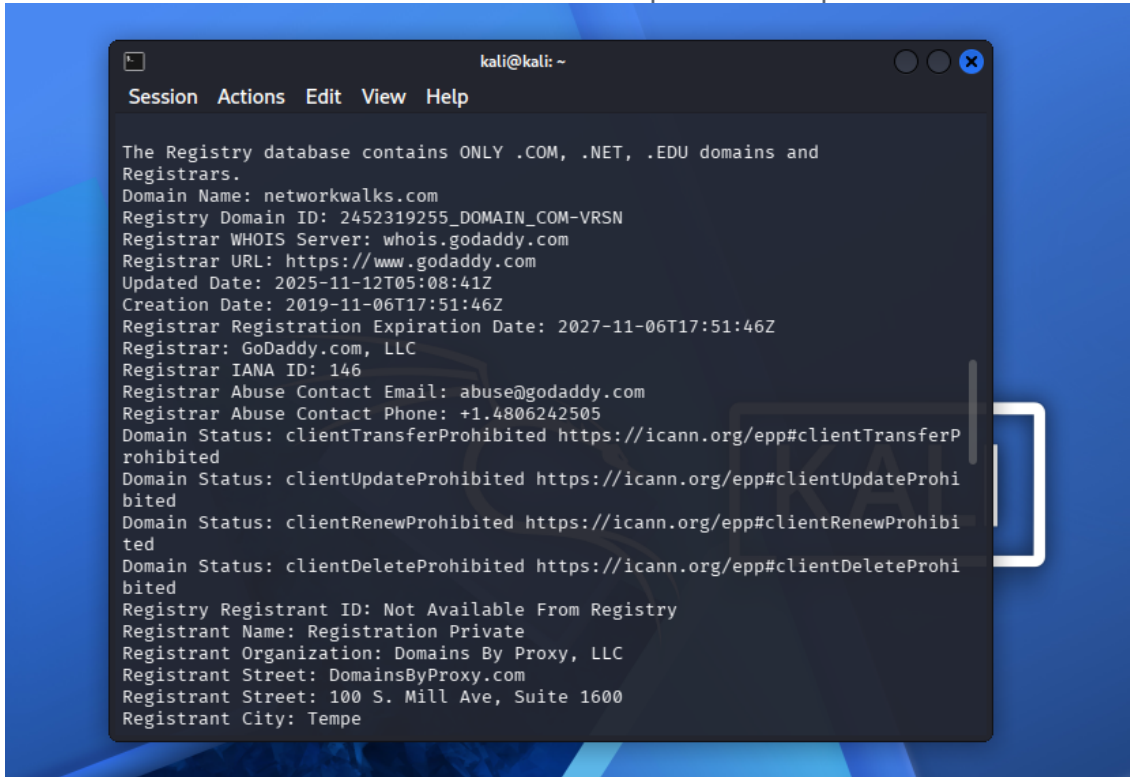


Figure 1. WHOIS evidence captured during the authorised footprinting exercise.

4.2 Web Technology Fingerprinting

WhatWeb identified a redirect from HTTP to HTTPS and a successful HTTPS response. The web stack presented Apache, WordPress 7.0.4, WordPress Download Manager 3.3.58, Bootstrap 7.0.4, jQuery 3.7.1, Google Tag Manager, HTML5, and Open Graph metadata. The output also exposed the website title, a public contact address, and an IP address associated with the response.

whatweb networkwalks.com

Category	Observed result
Transport behaviour	HTTP returned 301 Moved Permanently and redirected to HTTPS; HTTPS returned 200 OK.
Web server	Apache
CMS and component	WordPress 7.0.4; WordPress Download Manager 3.3.58
Front-end indicators	Bootstrap 7.0.4, jQuery 3.7.1, HTML5, Google Tag Manager, and Open Graph Protocol
Public metadata	Networkwalks Academy title and info@networkwalks.com address
Cookie indicator	__wpdm_client, observed with HttpOnly in the WhatWeb output

The principal security consideration is technology disclosure. Public version information can help a defender confirm patch levels, but it can also help an authorised tester compare the exposed stack with current security advisories. The result should not be interpreted as confirmation that the identified software is vulnerable.

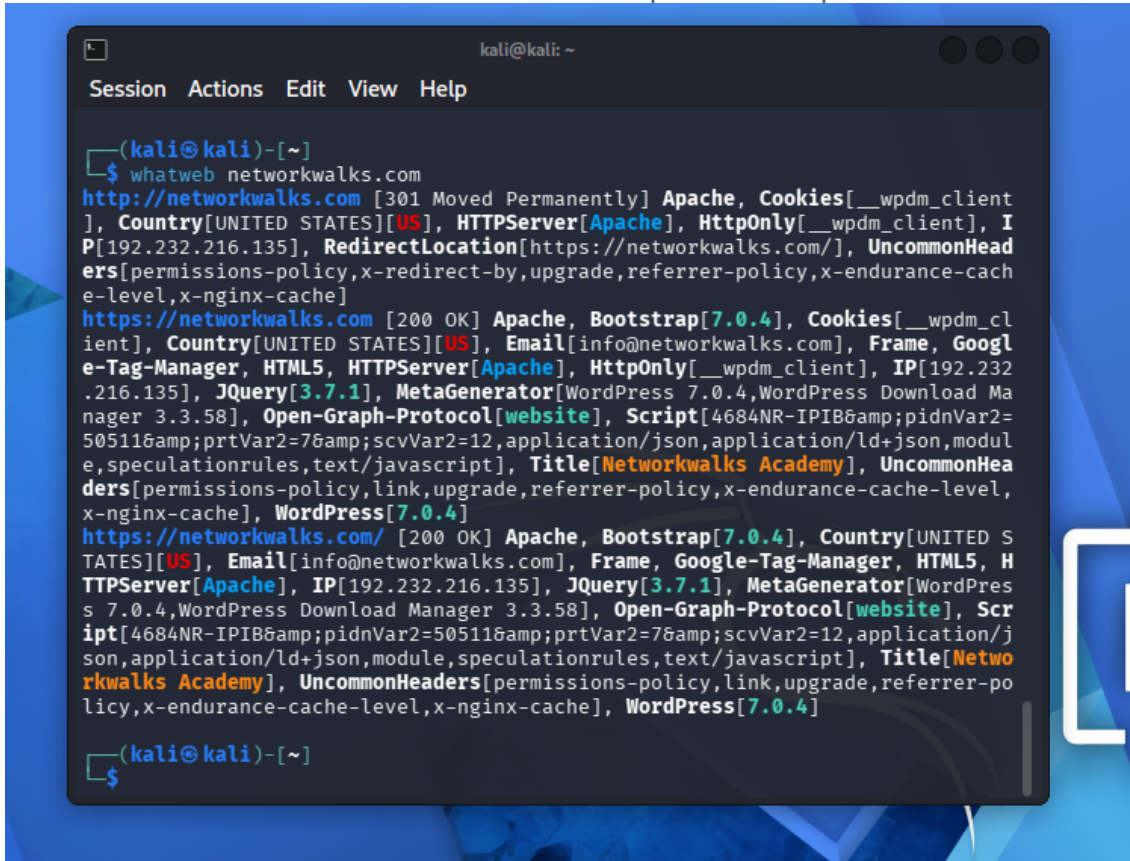


Figure 2. WhatWeb output showing the detected web technologies and metadata.

4.3 DNS Resolution

The nslookup query used Google Public DNS at 8.8.8.8 and returned a non-authoritative answer for networkwalks.com. The domain resolved to 192.232.216.135. The non-authoritative label indicates that the answering resolver was not the authoritative source of the domain zone; it does not, by itself, indicate that the response is incorrect.

```
nslookup networkwalks.com
```

DNS observation	Result
Resolver	8.8.8.8
Resolver port	53
Response type	Non-authoritative answer
Domain	networkwalks.com
Observed IPv4 address	192.232.216.135

The address is the public web-service address observed during the exercise. It should be treated as time specific because hosting arrangements, proxies, caching, and DNS changes can alter the answer.

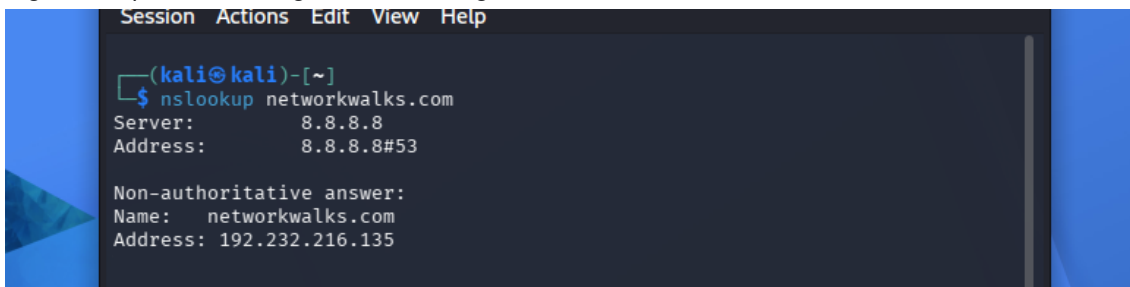


Figure 3. nslookup evidence showing the resolver and returned IPv4 address.

4.4 HTTP Response Header Inspection

The curl request returned HTTP/2 200, confirming that the target responded successfully to the request. The headers identified Apache, a WordPress-related cookie named __wpdm_client with Secure and HttpOnly attributes, the x-nginx-cache value WordPress, and a Link header exposing the WordPress REST API root and a page endpoint.

```
curl -I https://networkwalks.com
```

Header area	Observed result
Status	HTTP/2 200
Server	Apache
Content type	text/html; charset=UTF-8
Cookie	__wpdm_client with Secure and HttpOnly attributes
Cache indicators	x-endurance-cache-level: 0 and x-nginx-cache: WordPress
Referenced endpoints	/wp-json/ and /wp-json/wp/v2/pages/53
Policy header	referrer-policy: no-referrer-when-downgrade

The REST API references are not evidence of a vulnerability. They are public application metadata and should be reviewed by the owner to confirm that the exposed endpoints return only intended information. The cookie attributes observed in the output are positive controls for transport and script access, although the wider session-management design would require a separate review.

```

kali@kali: ~
Session Actions Edit View Help

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

(kali@kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=0da084ff4c8a74a650ce30aa69d3ec66; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Wed, 19 Aug 2026 11:24:41 GMT
server: Apache
  
```

Figure 4. curl header evidence from the HTTPS response.

4.5 Web Application Firewall Identification

wafw00f identified the site as being behind a ModSecurity Web Application Firewall associated with SpiderLabs. The tool reached this conclusion after two requests. This is an identification result based on observable response behaviour and signatures, not a guarantee that every malicious request would be blocked.

```
wafw00f networkwalks.com
```

WAF observation	Result
Tool version	WAFW00F v2.4.2
Target checked	https://networkwalks.com
Detected protection	ModSecurity (SpiderLabs) WAF
Requests used	2

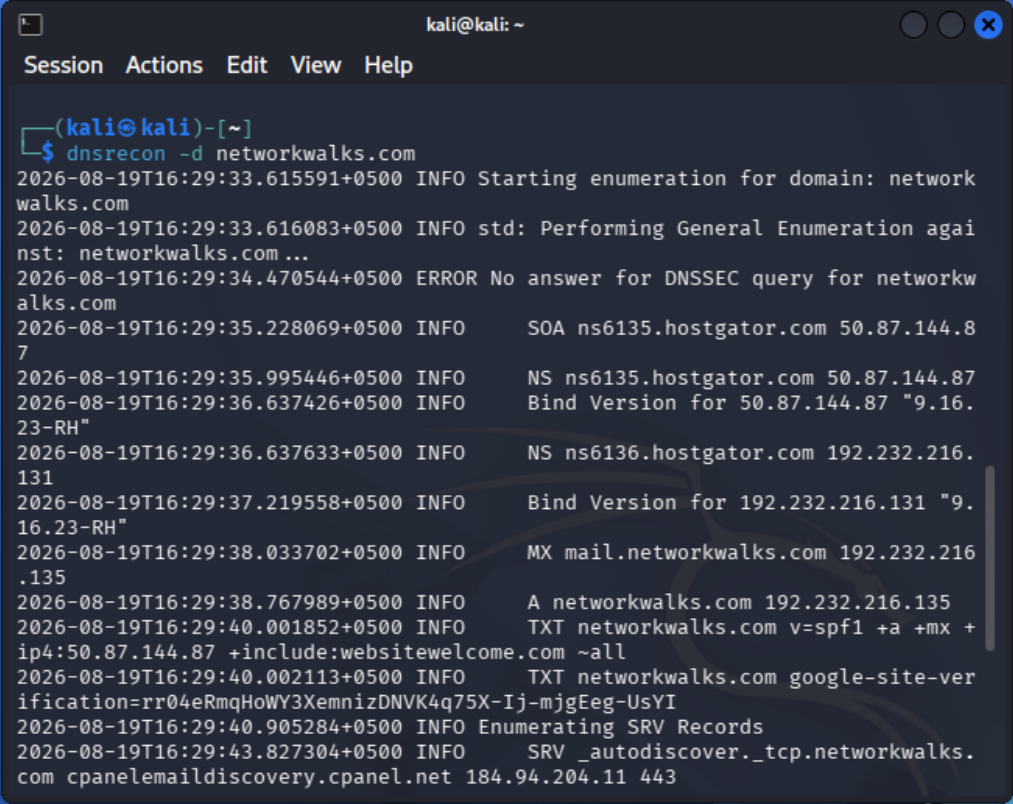
The presence of a WAF is a positive defensive control, but its effectiveness depends on rule quality, update frequency, monitoring, and correct placement in front of the application. A WAF should complement secure development and patch management rather than replace them.

4.6 DNS Record Enumeration

```
dnsrecon -d networkwalks.com
```

DNS category	Observed result
SOA and primary name server	ns6135.hostgator.com at 50.87.144.87
Secondary name server	ns6136.hostgator.com at 192.232.216.131
DNS software indicator	BIND 9.16.23-RH reported for both observed name-server addresses
Mail exchange	mail.networkwalks.com at 192.232.216.135
A record	networkwalks.com at 192.232.216.135
SPF record	v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
TXT record	Google site-verification token observed
SRV records	Eight _autodiscover._tcp records pointing to cpanelmaildiscovery.cpanel.net on port 443
DNSSEC query	No answer returned for the DNSSEC query

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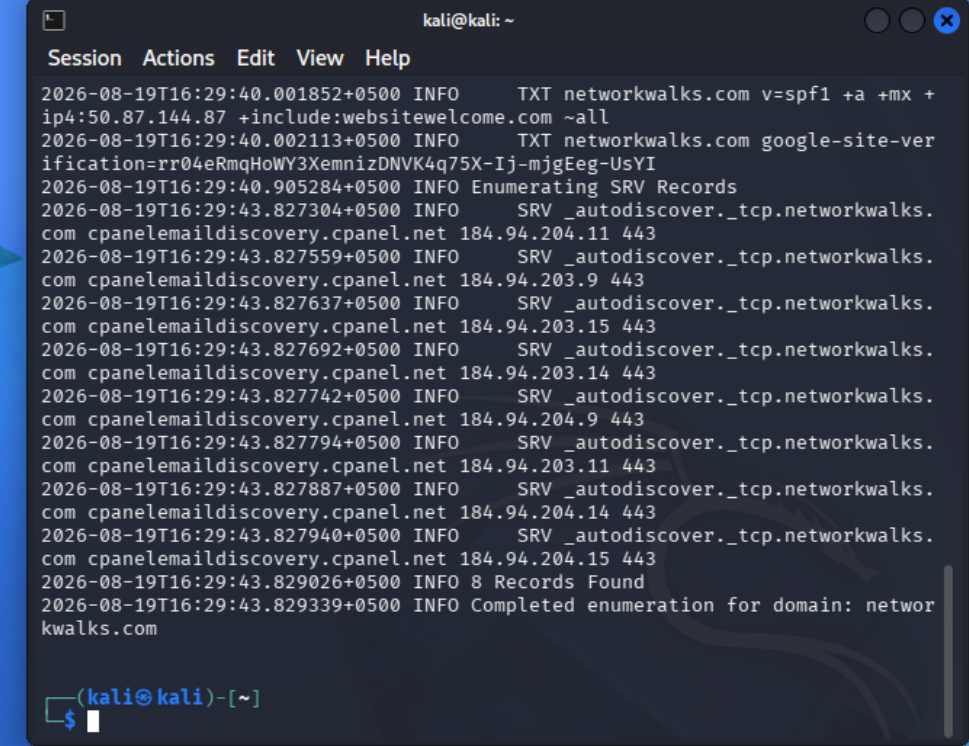
```

kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ dnsrecon -d networkwalks.com
2026-08-19T16:29:33.615591+0500 INFO Starting enumeration for domain: networkwalks.com
2026-08-19T16:29:33.616083+0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-08-19T16:29:34.470544+0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-08-19T16:29:35.228069+0500 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-08-19T16:29:35.995446+0500 INFO NS ns6135.hostgator.com 50.87.144.87
2026-08-19T16:29:36.637426+0500 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-08-19T16:29:36.637633+0500 INFO NS ns6136.hostgator.com 192.232.216.131
2026-08-19T16:29:37.219558+0500 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-08-19T16:29:38.033702+0500 INFO MX mail.networkwalks.com 192.232.216.135
2026-08-19T16:29:38.767989+0500 INFO A networkwalks.com 192.232.216.135
2026-08-19T16:29:40.001852+0500 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-08-19T16:29:40.002113+0500 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-08-19T16:29:40.905284+0500 INFO Enumerating SRV Records
2026-08-19T16:29:43.827304+0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443

```

Figure 6. DNSRecon evidence showing DNS, mail, TXT, SPF, and service records.



```

kali@kali: ~
Session Actions Edit View Help

2026-08-19T16:29:40.001852+0500 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-08-19T16:29:40.002113+0500 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-08-19T16:29:40.905284+0500 INFO Enumerating SRV Records
2026-08-19T16:29:43.827304+0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-08-19T16:29:43.827559+0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-08-19T16:29:43.827637+0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-08-19T16:29:43.827692+0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-08-19T16:29:43.827742+0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-08-19T16:29:43.827794+0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-08-19T16:29:43.827887+0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-08-19T16:29:43.827940+0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-08-19T16:29:43.829026+0500 INFO 8 Records Found
2026-08-19T16:29:43.829339+0500 INFO Completed enumeration for domain: networkwalks.com

(kali@kali)-[~]
$

```

Figure 7. Additional DNSRecon evidence captured for the enumeration task.

5. Network Scanning with Zenmap

The second exercise was conducted on an authorised local network using Zenmap, the graphical interface for Nmap. The scan used the Ping Scan profile. The captured result shows Nmap 7.991 scanning 256 addresses and identifying four hosts as up. The exact command shown in the evidence used 192.168.18.1/24; for clarity, the corresponding /24 network is reported separately as 192.168.18.0/24.

```
nmap -sn 192.168.18.1/24
```

Scan element	Observed result
Scanner	Zenmap / Nmap 7.991
Profile	Ping scan
Executed target	192.168.18.1/24, exactly as shown in the captured evidence
Normalised network	192.168.18.0/24 when expressed as a /24 network address
Address space scanned	256 IP addresses
Live hosts	4 hosts up
MAC evidence	MAC addresses were displayed in the scan output; values are redacted in this report copy
Topology	Zenmap topology view captured with the legend enabled

The scan was a host-discovery exercise rather than a vulnerability scan. It established which devices responded on the authorised local subnet and provided a visual topology representation. The public-report screenshot has been re-redacted so that the target subnet, private addresses, hardware identifiers, and device names are not exposed.

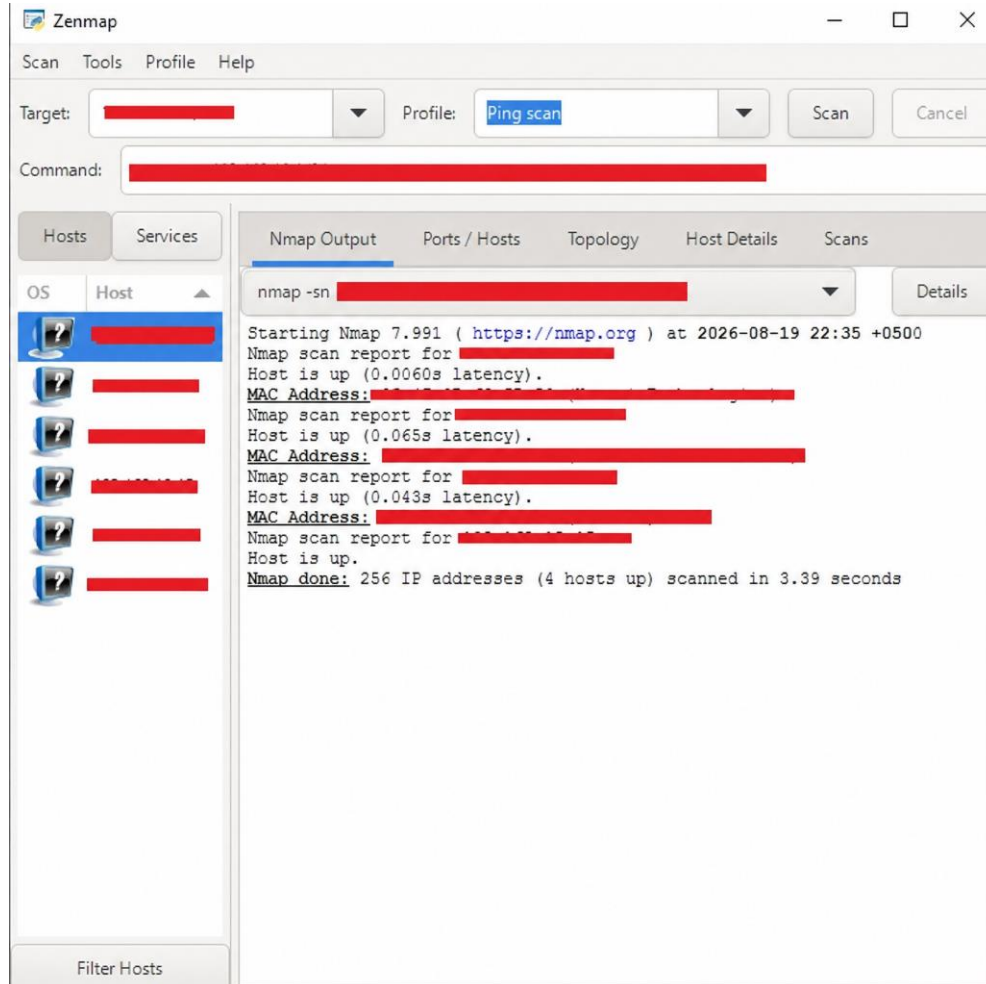


Figure 8. Re-redacted Zenmap Ping Scan output showing four live hosts while concealing the target subnet and device identifiers.

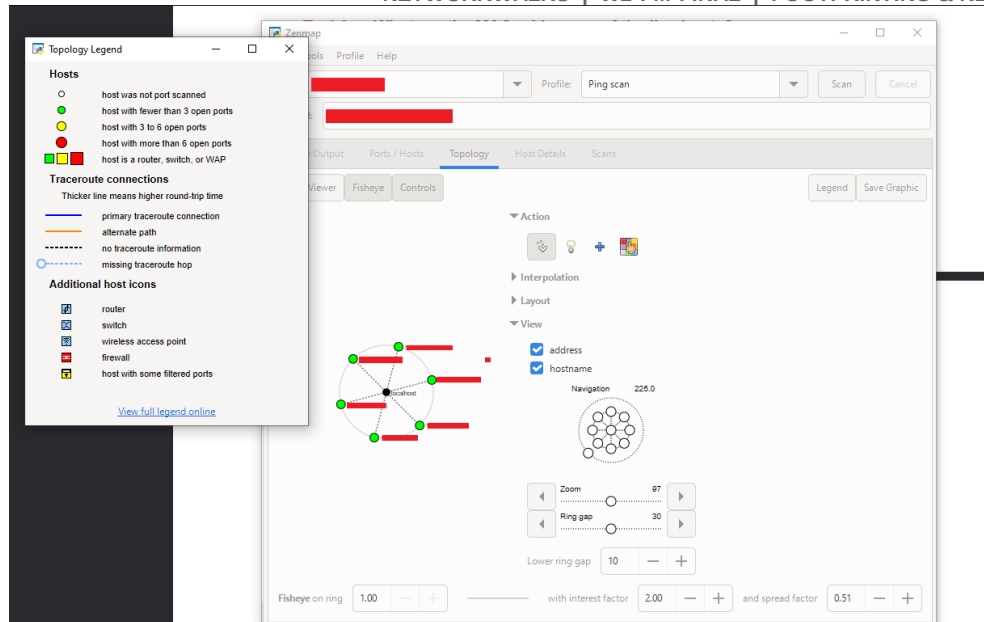


Figure 9. Redacted Zenmap topology view with the legend enabled.

Privacy control applied

The source screenshots contain redactions over local IP addresses, MAC addresses, host labels, and the target subnet in the public-report copy. The original evidence should be retained separately for private academic submission.

6. Risk and Impact Analysis

The following table records security observations arising from the exercises. The ratings describe the potential value of the information to a future attacker, not a confirmed vulnerability or compromise.

ID	Finding	Evidence	Potential impact	Rating
F-01	Web stack and component versions visible	WhatWeb identified WordPress 7.0.4, WordPress Download Manager 3.3.58, Bootstrap 7.0.4, jQuery 3.7.1, and Apache.	May assist version correlation and targeted security review.	Medium
F-02	Public web address resolved	nslookup and DNSRecon returned 192.232.216.135 for the website.	Adds infrastructure context and supports further authorised assessment planning.	Low
F-03	HTTP metadata and application links exposed	curl returned server, cache, cookie, and WordPress REST API link headers.	May assist technology fingerprinting and application mapping.	Low
F-04	WAF technology identifiable	wafw00f identified ModSecurity (SpiderLabs).	Reveals part of the defensive architecture, although it is also a positive control.	Low
F-05	DNS and service footprint exposed	DNSRecon identified name servers, mail routing, SPF/TXT records, SRV records, cPanel infrastructure, and BIND version indicators.	Supports broader infrastructure profiling and configuration review.	Medium
F-06	Multiple local hosts responded	Zenmap identified four live hosts on the authorised /24 subnet.	Unknown devices could represent unmanaged or unauthorised assets if not reconciled with an inventory.	Medium

Assessment boundary

None of these observations confirms a vulnerability. The work performed was limited to public-footprint collection and local host discovery. Validation would require separate authorisation, test objectives, and safeguards.

7. Recommendations

- 1. Reduce unnecessary web technology disclosure.** Review server banners, generator metadata, plugin fingerprints, and public contact details. Expose only information required for normal operation.
- 2. Maintain the WordPress stack.** Keep WordPress, plugins, themes, JavaScript libraries, and server components supported and updated. Remove unused components rather than leaving them installed.

3. Review HTTP security posture. Confirm that cookies, redirects, cache behaviour, transport security, and security headers match the application risk profile. Review the public WordPress REST API endpoints for intended disclosure.

4. Monitor the DNS footprint. Review A, MX, NS, TXT, SPF, and SRV records regularly. Remove stale records and restrict DNS changes through strong registrar controls and change alerts.

5. Protect and tune the WAF. Keep ModSecurity enabled, maintain current rules, monitor blocked traffic, and investigate false positives without weakening protection unnecessarily.

6. Reconcile live local hosts. Compare Zenmap results with an approved asset inventory. Investigate devices that are unknown, unnecessary, or not assigned to a responsible owner.

7. Protect assessment evidence. Keep original screenshots and raw command outputs in a private submission folder. Use redacted copies for LinkedIn, GitHub, or other public platforms.

8. Evidence Register

The evidence package is organised by task. Each item below can be traced to the raw command output or screenshot supplied for this report.

Evidence ID	Tool or activity	Supplied material	Purpose
Task 1	WHOIS	Networkwalks.txt and two screenshots	Registration, registrar, lifecycle, status, and name-server evidence
Task 2	WhatWeb	Whatweb.txt and screenshot	Web technology and metadata fingerprinting
Task 3	nslookup	nslookup.txt and screenshot	Resolver and IPv4 resolution evidence
Task 4	curl -I	-I Curl.txt and screenshot	HTTP response header evidence
Task 5	wafw00f	wafw00f.txt and screenshot	WAF identification evidence
Task 6	DNSRecon	dnsrecon.txt and two screenshots	DNS, mail, TXT, SPF, and SRV evidence
Zenmap	Ping Scan and Topology	Redacted scan screenshot and redacted topology screenshot	Four live hosts and visual topology evidence

9. Conclusion

The two Week 2 exercises demonstrated the relationship between reconnaissance and network discovery. The footprinting work showed how registration data, web responses, DNS records, application metadata, and WAF signatures can be combined into a meaningful public profile. The Zenmap work showed how a controlled host-discovery scan can identify active devices and present the local network in a topology view.

The most important lesson is that the collection of small technical details can create a larger operational picture. Technology versions can be correlated with security advisories, DNS records can expose hosting and mail relationships, HTTP headers can reveal application paths, and an unmanaged live host can become a blind spot in an internal network.

The exercises were completed within an educational and authorised scope. No exploitation or vulnerability validation was attempted. The next professional step would be to validate the observations with the asset owner, confirm patch and configuration status, and repeat the checks periodically so that changes in the public footprint or local network are detected early.

Appendix A. Commands Executed

```
whois networkwalks.com
```

```
whatweb networkwalks.com
```

```
nslookup networkwalks.com
```

```
curl -I https://networkwalks.com
```

```
wafw00f networkwalks.com
```

```
dnsrecon -d networkwalks.com
```

```
nmap -sn 192.168.18.1/24 [executed command; /24 network normalised as 192.168.18.0/24]
```

Appendix B. Source References

Ref.	Source	URL
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[1]	Nmap Project, Nmap Reference Guide, Host Discovery	https://nmap.org/book/man-host-discovery.html
[2]	Nmap Project, Zenmap GUI Documentation	https://nmap.org/zenmap/
[3]	IETF RFC 1035, Domain Names: Implementation and Specification	https://datatracker.ietf.org/doc/html/rfc1035
[4]	Microsoft Learn, nslookup command	https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/nslookup
[5]	OWASP, Web Application Security Testing	https://owasp.org/www-project-web-security-testing-guide/

End of Report

Prepared for educational cybersecurity assessment purposes